

Traffic Light Recognition

This game involves using commands to enable the camera to recognize traffic lights.

1. Getting Ready

- 1) Before starting, ensure the map is laid out flat and free of wrinkles, with smooth roads and no obstacles. For specific map laying instructions, please refer to "Map Laying and Prop Installation" in the same directory as this section for guidance.
- 2) The roadmap model discussed in this section is trained using YOLOv5. For further information on YOLOv5 and related content, please refer to "16. Machine Learning".
- 3) When experiencing this game, ensure it is conducted in a well-lit environment, but avoid direct light shining on the camera to prevent misrecognition.

2. Program Logic

Firstly, capture a real-time image from the camera and apply operations such as erosion and dilation.

Next, invoke the YOLOv5 model to compare it with the target screen image.

Finally, execute corresponding landmark actions based on the comparison results.

The source code for this program can be found at:

```
/home/ubuntu/ros2_ws/src/example/example/yolov5_detect/yolov5_trt.py
```

3. Operation Steps

1. Note: The following steps exclusively activate road sign detection in the return screen, without executing associated actions. Users seeking direct experience with

autonomous driving may bypass this lesson and proceed to "Integrated Application" within the same file.

2. When inputting commands, please ensure accurate differentiation between uppercase and lowercase letters as well as spaces. Utilize the "Tab" key for keyword completion.

1) Start the robot, and access the robot system desktop using VNC.



2) Click-on  to open the command-line terminal.

3) Execute the command to disable the app auto-start service.

```
~/stop_ros.sh
```

```
> ~/.stop_ros.sh
```

4) Run the command to open the camera node.

```
ros2 launch peripherals depth_camera.launch.py
```

```
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```

5) Open a new command line terminal. Enter the command to enable the camera node.

```
ros2 launch yolov5_ros2 yolov5_ros2.launch.py
```

```
model:="traffic_signs_640s_7_0"
```

```
ros2 launch yolov5_ros2 yolov5_ros2.launch.py model:="traffic_signs_640s_7_0"
```

6) Open a new command line terminal. Execute the command to open the interface for live camera feed.

```
rqt
```

```
> rqt
QStandardPaths: XDG_RUNTIME_DIR not set, defaulting to '
tmp/runtime-ubuntu'
```

7) Position the road sign in front of the camera, and the robot will recognize the road sign and mark it on the image. If you need to terminate this game, press "Ctrl+C". If it fails, please try again.

Note: In the event that the model struggles to recognize traffic-related signs, it may be necessary to lower the confidence level. Conversely, if the model

consistently misidentifies traffic-related signs, raising the confidence level might be advisable.

8) Run the command to navigate to the directory containing programs.

```
cd /home/ubuntu/ros2_ws/src/example/example/self_driving
```

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cd /home/ubuntu/ros2_ws/src/example/example/self_driving
```

9) Execute the command to open the game program.

```
vim self_driving.launch.py
```

```
vim self_driving.launch.py
```

The red box represents the confidence value, which can be adjusted to modify the effectiveness of target detection.

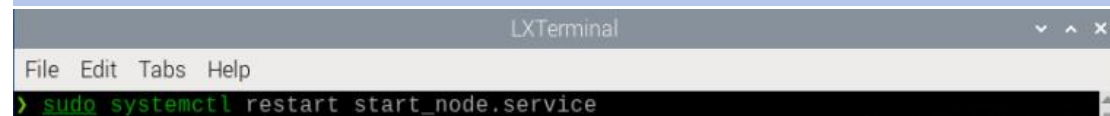
```
80 )
81
82 yolo5_node = Node(
83     package='example',
84     executable='yolo5_node',
85     output='screen',
86     parameters=[{'classes': ['go', 'right', 'park', 'red', 'green', 'crosswalk']},
87                 {'use_depth': 'true', 'engine': 'traffic_signs_640s_7_0.engine', 'lib': 'libmyplugins.so', 'conf': 0.75}],
88 )
89
90 self_driving_node = Node(
91     package='example',
92     executable='self_driving',
93     output='screen',
94     parameters=[{'start': start}, {'only_line_follow': only_line_follow}],
95 )
```

Once you've completed the game experience, you can initiate the app service by following the instructions or restarting the robot.

If the app service is not activated, the associated app functions will be disabled. (The app service will automatically start when the robot restarts).

To restart the app service, enter the following command and wait for the buzzer to emit a single beep, indicating that the service startup is complete.

```
sudo systemctl restart start_node.service
```



4. Program Outcome

After initiating the game, position the robot on the road depicted on the map.

Upon recognizing the traffic signal, the robot will assess the color of the signal

light and identify frames corresponding to red and green signal lights accordingly.

